

CS403 CURRENT SOLVED SUBJECTIVE MID PAPERS BY SARIM

Database Management Systems

Prepared by **Sarim**

Q1) Procedural and nonprocedural language difference

Answer:

Procedural languages require explicit instructions on how to accomplish a task, while nonprocedural languages specify what needs to be done without detailing how to do it.

Examples of procedural languages include C, Pascal, and COBOL, while nonprocedural languages include SQL and HTML.

Q2) Application development process

Answer:

Application development involves several stages including requirements gathering, design, implementation, testing, deployment, and maintenance.

Each stage requires careful planning and execution to ensure the final application meets the needs of users.

Q3) Relational algebra properties

Answer:

1. Closure
2. Associativity
3. Commutativity
4. Idempotence
5. Distributivity

Q4) 3 level architecture men use hone wali mapping ki types

Answer:

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1. Internal-Internal
2. Internal-Conceptual
3. Conceptual-External

Q5) Foreign keys and home relation

Answer:

- A foreign key is a column or a set of columns in a database table that refers to the primary key or a unique key in another table.
- Home relation refers to the table in which the referenced primary key or unique key resides.

Q6) Total relationship and partial relationship with real life example

Answer:

- Total Relationship: Every entity in the entity set must participate in the relationship. Example: A classroom must have at least one student enrolled.
- Partial Relationship: Some entities in the entity set may not participate in the relationship. Example: A student may or may not be enrolled in any classroom.

Q7) multi-valued attribute

Answer:

A multi-valued attribute can have multiple values for a single entity.

Example: Skills of an employee (e.g., programming, management, communication).

Q8) Advantages of database system

Answer:

- Data Sharing
- Data Integrity
- Data Security

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- Data Independence
- Efficient Data Access

Q9) Stored and derived attributes difference

Answer:

- Stored attributes are physically stored in the database, while derived attributes are calculated from other attributes.
- Stored attributes are directly represented in the database schema, while derived attributes are computed on-the-fly when needed.

Q10) Constraints

Answer:

Constraints are rules enforced on data columns to maintain data integrity and accuracy in the database. Common types of constraints include primary key, foreign key, unique, check, and default constraints.

Q11) Data redundancy

Answer:

Data redundancy refers to the duplication of data in multiple places within a database. It can lead to inconsistencies and inefficiencies in data management.

Q12) Type of attribute

Answer:

Types of attributes include single-valued attribute, multi-valued attribute, composite attribute, and derived attribute.

Q13) Unary operation?

Answer:

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A unary operation operates on a single operand. Example: Negation, Increment, Decrement

Q14) What is difference between strong and weak database management system (DBMS)

Answer:

Strong DBMS enforces referential integrity constraints, ensuring that foreign key values correspond to existing primary key values in related tables.

A weak DBMS does not enforce referential integrity constraints, allowing foreign key values to be null or refer to non-existent primary key values

Q15) Skills attribute is which type of attribute?

Answer:

Skills attribute is a multi-valued attribute.

Q16) Name 2 mapping schema?

Answer:

Two mapping schemas are Internal Schema and External Schema

Q17) In three level architecture which level is presenting information to end users?

Answer:

The external level presents information to end users.

Q18) one-to-many relationship draw krna tha between student and department

Answer:

Student -----< Department

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Q19) What is Preliminary Study.

Answer:

Preliminary study is the initial phase of the database development process where the feasibility of the project is evaluated, and requirements are gathered.

Q20) Equi join, theta join, outer join

Answer:

Equi Join: Joins two tables based on matching values in a specified column.

Theta Join: Join based on a condition other than equality.

Outer Join: Includes rows from one table that do not have a match in the other table.

Q21) Database maintenance system ko define krna tha?

Answer:

Database maintenance system involves activities related to ensuring the ongoing health, performance, and integrity of a database, including backups, updates, security measures, and performance tuning.

Q22) Schema, Application management, Database management ko explain krna tha

Answer:

- Schema: Describes the structure of the database, including tables, columns, and relationships.
- Application Management: Involves managing the software applications that interact with the database.
- Database Management: Involves tasks related to storing, organizing, and retrieving data in a database system.

Q23) Student

Course

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Program

Teacher

In ko kesy likhty hain wo btana tha

Answer:

- Student (1:N) Course
- Program (1:N) Course
- Teacher (N:1) Course

Q24) Database development process

Answer:

Database development process involves steps like requirements analysis, conceptual design, logical design, physical design, implementation, testing, deployment, and maintenance.

Q25) Natural join ka table bnaya

Answer:

StudentID | Name | CourseID | CourseName

101 | Alice | 1 | Math

101 | Alice | 2 | English

102 | Bob | 1 | Math

102 | Bob | 3 | Science

Q26) Difference between procedure language or non- procedure language

Answer:

Procedural languages require explicit instructions on how to accomplish a task, while non-procedural languages specify what needs to be done without detailing how to do it.

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Q27) Weak Aur strong entity type ka difference batana Tha

Answer:

A weak entity type depends on a strong entity type for its existence and is indicated by a double rectangle in the E-R diagram.

A strong entity type exists independently and is represented by a single rectangle.

Q28) 2 architecture levels ki definition di hoi thin level name likhna tha kh kis level ki definition h

Answer:

Two architecture levels in database systems are

- External level: It defines how individual users or user groups view the data.
- Conceptual level: It defines the overall logical structure of the entire database for the entire organization.

Q29) Semantic data model ka ilawa jo aik or data model tah oski types ka name likna tha

Answer:

Apart from the Semantic data model, another type of data model is the Object-oriented data model.

Q30) hierarchy and relational data model?

Answer:

Hierarchy data model organizes data into a tree-like structure, while the relational data model organizes data into tables with rows and columns.

Q31) Candidate key

Answer:

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Candidate key is a minimal superkey, meaning it uniquely identifies each record in a relation and does not contain unnecessary attributes.

Q32) Levels of data

Answer:

Levels of data include

1. Physical level
2. Logical level
3. View level

Q33) Write the three names of multi user DBMS environment?

Answer:

The three names of multi-user DBMS environments are

1. Teleprocessing
2. File server
3. Client-server

Q34) Difference between referential integrity and entity integrity constraints?

Answer:

Referential integrity constraints ensure that relationships between tables remain valid, while entity integrity constraints ensure that each row in a table is uniquely identifiable.

**Q35) Tabel is given match the columns and match the entities with their labels:
Entity Name; Mr Ahmed Entity set; Employee Entity instances; All Vu departments**

Answer:

- Entity Name: Employee
- Entity set: All Vu departments

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- Entity instances: Mr. Ahmed

Q36) Related to Cartesian Products

Answer:

Applying the same concept in a real world scenario, consider two sets Name and Age having the elements:

- Name = {Ali, Sana, Ahmed, Sara}
- Age = {15,16,17,18,.....,25}

Cartesian product of Name & Age

Name X Age = {(Ali,15), (Sana,15), (Ahmed,15), (Sara,15), ..., (Ahmed,25), (Sara,25)}

Now consider a subset CLASS of this Cartesian product

CLASS = {(Ali, 18), (Sana, 17), (Ali, 20), (Ahmed, 19)}

Q37) Redundant data explain

Answer:

Redundant data refers to the duplication of data in multiple places within a database, leading to inconsistencies and inefficiencies in data management.

Q38) Difference b/w Database System and Traditional File Processing

Answer:

In database systems, data is organized centrally and accessed through a database management system, whereas in traditional file processing, data is stored in separate files, leading to data redundancy and inconsistency.

Q39) Database architecture layers.

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Answer:

Database architecture layers include the external level (user views), conceptual level (logical schema), and internal level (physical storage).

Q40) data model ke types

Answer:

Types of data models include hierarchical, network, relational, object-oriented, and entity-relationship models.

Q41) Five basic operation of relational algebra

Answer:

1. Selection
2. Projection
3. Union
4. Set Difference
5. Cartesian Product.

Q42) Functional dependencies

Answer:

Functional dependencies describe the relationships between attributes in a relation. An example is if A determines B, denoted as $A \rightarrow B$.

Q43) Entity and its types

Answer:

An entity is a real-world object or concept with an independent existence. Types include strong and weak entities.

Q44) List the names of Database development process

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Answer:

The names of the stages in the database development process include

- Requirement Analysis
- Implementation
- Testing
- Deployment
- Maintenance.

Q45) Super/Sub-Types

Answer:

Super-types are general categories, and sub-types are specific categories that inherit attributes from super-types. For example, Vehicle is a super-type, and Car and Truck are sub-types.

Q46) Context Level Diagram

Answer:

A context level diagram provides an overview of the system, showing external entities interacting with the system and the major processes within the system.

Q47) CRM (Cross Reference Matrix)

Answer:

A CRM is a tool used in project management to track requirements and ensure they are implemented correctly. It maps requirements to specific components or modules of the system.

Q48) E-R Diagram-Maximum and Minimum Cardinality (Student, Bus & Book)

Answer:

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Maximum and minimum cardinality constraints describe the minimum and maximum number of instances of one entity that can be associated with another entity in a relationship

Q49) Database monitoring per or is ka sath disk data adminstration tha different

Answer:

Database monitoring involves tracking the performance and health of a database system, while disk data administration involves managing storage devices and optimizing disk usage.

Q50) Write entities 6 entities for ERD on your University Examination system

Answer:

Student, Course, Faculty, Department, Exam, Result

Q51) Define domain of an attribute. Write it's significance. Also write one Example

Answer:

The domain of an attribute defines the set of allowable values for that attribute.

For example, the domain of the attribute "Age" may be integers between 0 and 150. It ensures data integrity and consistency.

Q52) Two inference rule are given write their names.

Answer:

Two inference rules are Armstrong's axioms and transitive rule.

Q53) What is the difference between strong entity and weak entity

Answer:

A strong entity exists independently and has a primary key, while a weak entity depends on another entity for identification and does not have a primary key of its own.

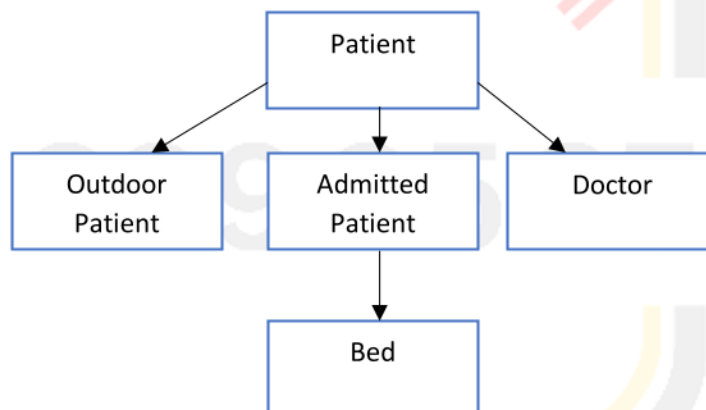
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Q54) Draw an ERD of patient management system on the basis of information given below

The entities involved in patient management system are as under.

- i) Patient
- ii) Outdoor Patient
- iii) Admitted Patient
- iv) Doctor
- v) Bed

Answer:



Q55) Difference between Primary key and alternative key 2 points

Answer:

- Primary Key: Unique identifier for each record in a table, cannot contain NULL values, and can be composite.
- Alternative Key: Candidate key that is not selected as the primary key, can also uniquely identify each record, and can be used as a foreign key in other tables

Q56) If organization is using file server how you can agree a higher management to use a DBMS brief

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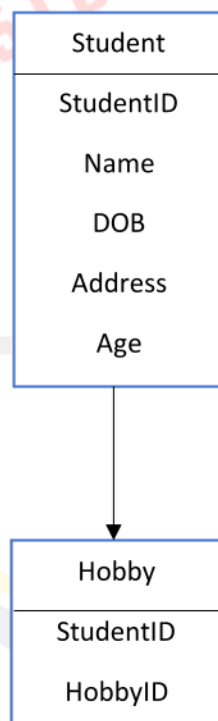
Answer:

The benefits of a DBMS such as data integrity, security, data consistency, centralized management, support for concurrent access, and data independence.

Emphasize the advantages in terms of efficiency, scalability, and ease of data retrieval and manipulation.

Q57) Scenario diya tha student ka. Student id name date of birth address age In ki ERD Diagram bnani thi Student id name ko composite Hobby ko multivalued m

Answer:



Q58) Standardization ka benefit btana tha future m apny point of view s

Answer:

Standardization ensures uniformity and consistency in processes, data formats, and communication protocols across different systems and organizations. This facilitates

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interoperability, reduces errors, improves efficiency, and enables seamless integration of systems.

Q59) Properties of relation

Answer:

Atomicity, Unique column names, Attribute domains, Null values, Tuple-based, Unordered, and No duplicate tuples.

Q60) how data is store in os physical level

Answer:

Data is stored in blocks or pages on secondary storage devices such as hard drives. Each block has a unique address and can store a fixed amount of data. The operating system manages the allocation and retrieval of data from these blocks.

Q61) properties of context level diagram

Answer:

Context level diagrams provide an overview of the entire system, showing external entities interacting with the system boundary and the major processes within the system. They emphasize the system's boundaries and interfaces with external entities.

Q62) Primary keys

Answer:

Primary keys uniquely identify each record in a table and ensure data integrity by enforcing entity integrity constraints. They must be unique and not contain NULL values.

Q63) 3 disadvantages of file processing system

Answer:

- ✚ Data redundancy and inconsistency
- ✚ Lack of data sharing and integration

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- ✚ Difficulty in data retrieval and manipulation
- ✚ Limited data security and integrity controls
- ✚ Lack of data independence and flexibility

Q64) Alternative and secondary key ki definition

Answer:

- Alternative Key: Candidate key that is not selected as the primary key but can uniquely identify each record in a table.
- Secondary Key: Non-primary key attribute that is used for data retrieval and indexing purposes but may not uniquely identify each record

Q65) Requirements analysis phase

Answer:

Requirements analysis phase is a stage in the software development process where the needs and expectations of stakeholders are gathered, documented, and analyzed to define the scope and objectives of the project.

Q66) Difference between file processing and database system

Answer:

File processing involves storing data in separate files using application programs, while a database system stores data centrally in a structured format with the help of a Database Management System (DBMS).

Q67) Difference btw intension and extension of the database

Answer:

Intension of the database refers to the schema or structure that defines the database objects and their relationships, while extension refers to the actual data stored in the database.

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Q68) Role of running job or administrator mein difference

Answer:

The running job performs specific tasks within a system, while an administrator manages the overall system and its resources

Q69) Types of join.

Answer:

Types of joins include inner join, outer join (left, right, and full), cross join, and self join.

Q70) Advantages of data base

Answer:

Advantages of a database include data consistency, data integrity, data security, efficient data retrieval, and data sharing.

Q71) Products, products range, product object write Anse one to many mapping

One to many

Many to one

Many to many

One to one

Answer:

- One-to-many mapping: Each product has multiple product range entries.
- Many-to-one mapping: Multiple product range entries belong to one product.
- Many-to-many mapping: Multiple products can belong to multiple product ranges.
- One-to-one mapping: Each product corresponds to only one product object.

Q73) What is date model? Write down list.

Answer:

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Data model defines the structure, relationships, constraints, and operations for organizing and accessing data. Types of data models include hierarchical, network, relational, object-oriented, and NoSQL.

Q74) Data dependency definition and as ki type name and ... Ak type ko explain Karna tha

Answer:

Data dependency refers to the relationship between data elements in a database. Types of data dependencies include functional dependency, multivalued dependency, and join dependency.

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